SFU NatLangLab

CMPT 413/713: Natural Language Processing

Log-linear models

Fall 2021
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Adapted from slides from Danqi Chen and Karthik Narasimhan
(with some content from Anoop Sarkar)

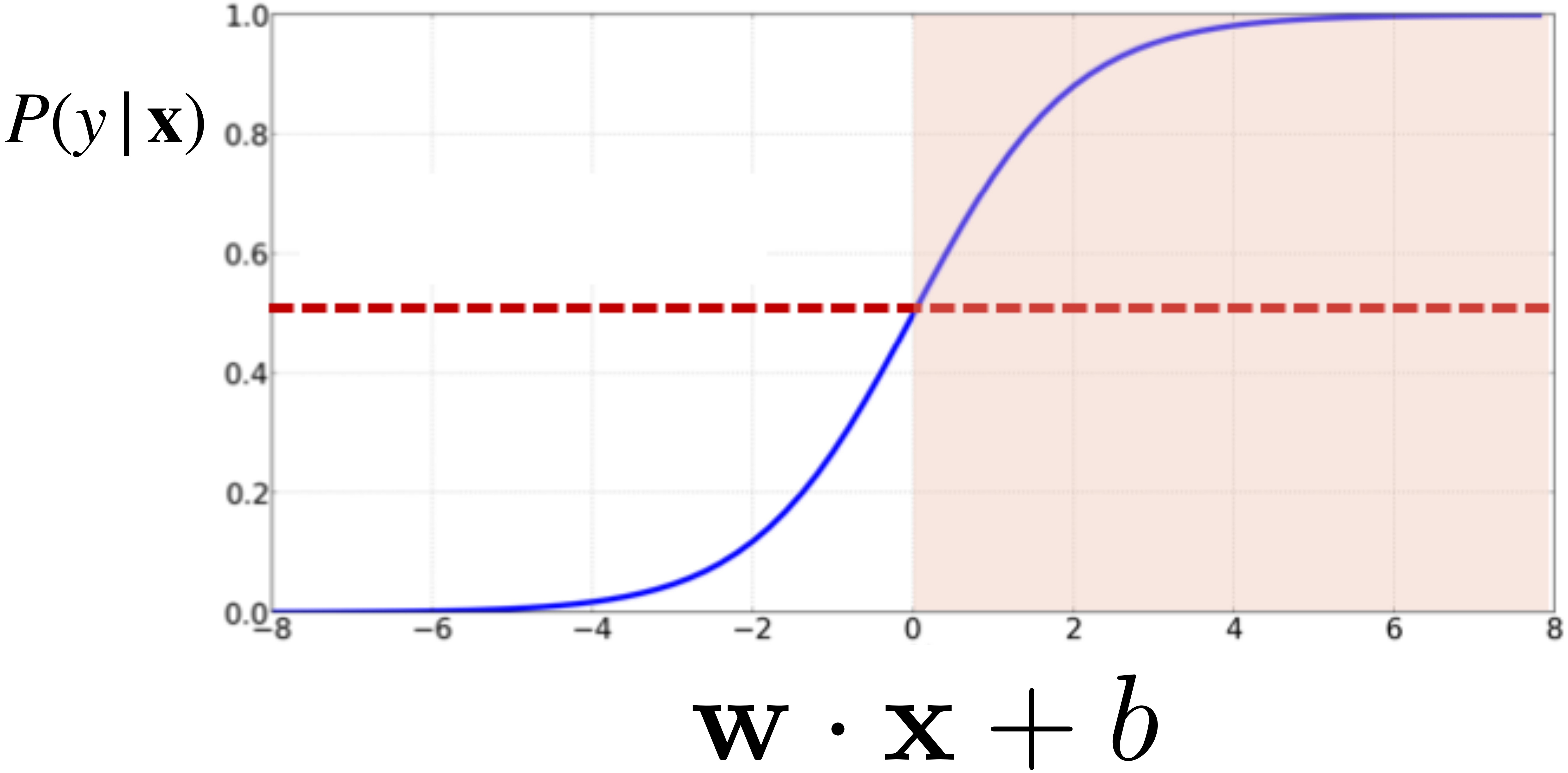
Last time

- Supervised classification:
  - Document to classify, d
  - Set of classes, $C = \{c_1, c_2, \ldots, c_k\}$
- Naive Bayes:

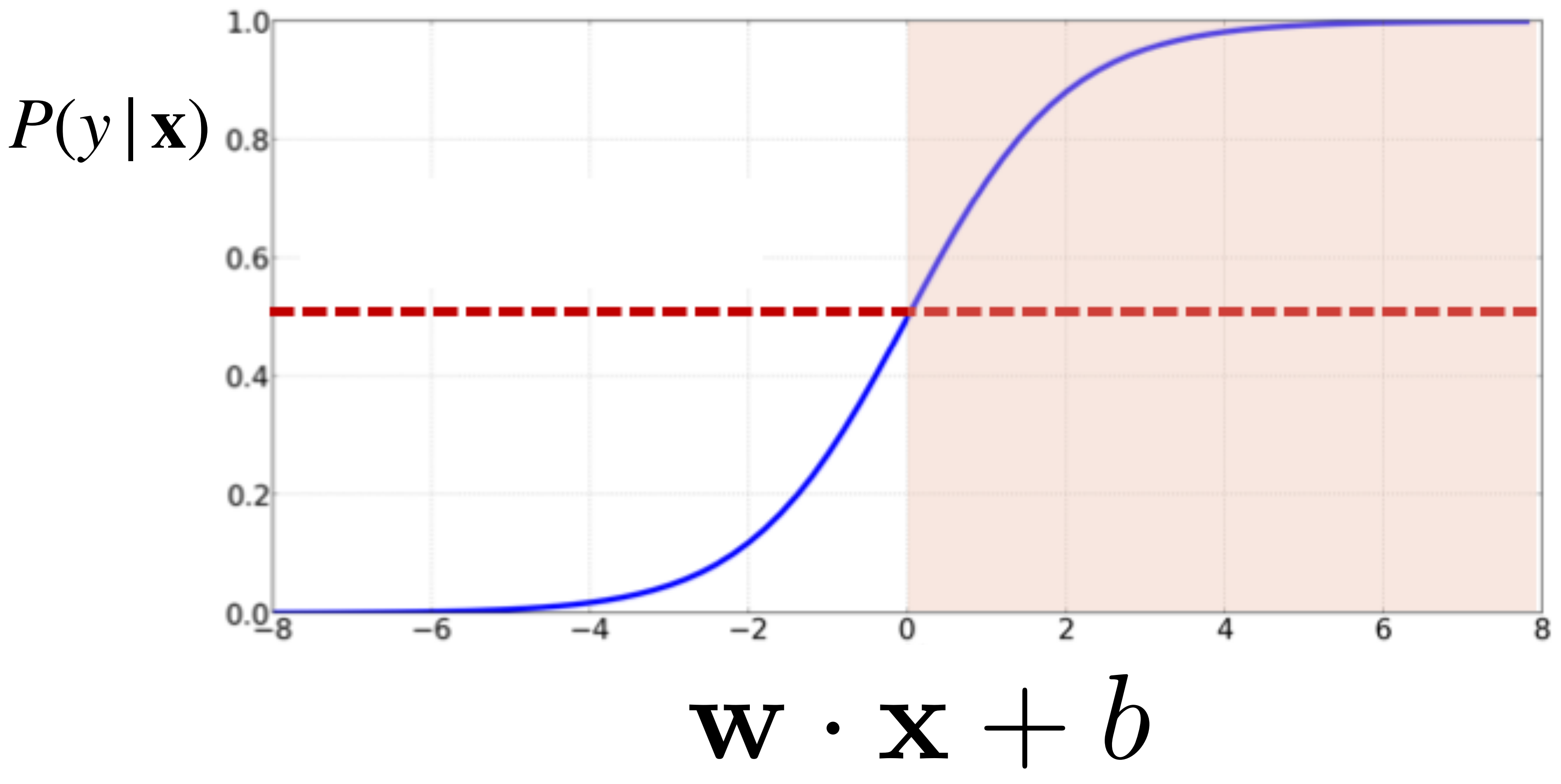
$$\hat{c} = \arg\max_{c} P(c)P(d \mid c)$$

Discriminative Model

- Logistic Regression: model $p(y \mid x)$ directly



Logistic Regression



- Powerful supervised model
- Baseline approach to most NLP tasks
- Connections with neural networks
- Binary (two classes) or multinomial (>2 classes)

Discriminative Model

- Logistic Regression is a *discriminative* model

  focus on discriminating features

- Naive Bayes: *generative* model

  can be used to generate the cat and the dog

cats have whiskers

Discriminative Model

- Logistic Regression: $\hat{c} = \underset{c}{\arg\max}\, P(c \mid d)$
- Naive Bayes: $\hat{c} = \underset{c}{\arg\max}\, P(c)P(d \mid c)$

Using Logistic Regression

- Inputs:

  1. Classification instance in a **feature representation**

  2. **Classification function** to compute $\hat{y}$ using $P(\hat{y} \mid x)$

  3. **Loss function** (for learning)

  4. Optimization **algorithm**

Training objective

- Train phase: Learn the **parameters** of the model to minimize **loss function**

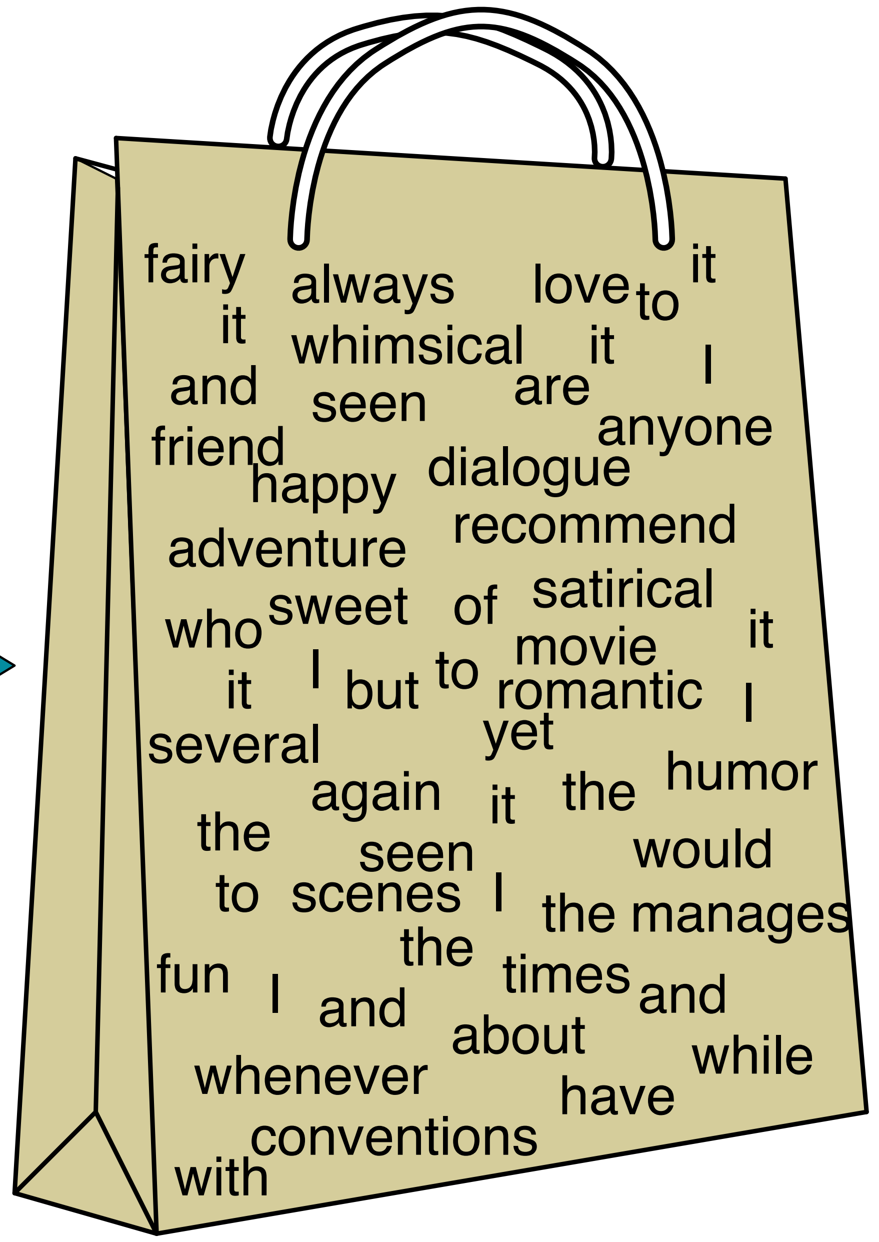
- Test phase: Apply **parameters** to predict class given a new input x

Feature representation

- Input observation: $x^{(i)}$
- Feature vector: $[x_1, x_2, \ldots, x_d]$
- Feature j of ith input : $x_j^{(i)}$

I love this movie! It's sweet, but with satirical humor. The dialogue is great and the adventure scenes are fun... It manages to be whimsical and romantic while laughing at the conventions of the fairy tale genre. I would recommend it to just about anyone. I've seen it several times, and I'm always happy to see it again whenever I have a friend who hasn't seen it yet!

Bag of words



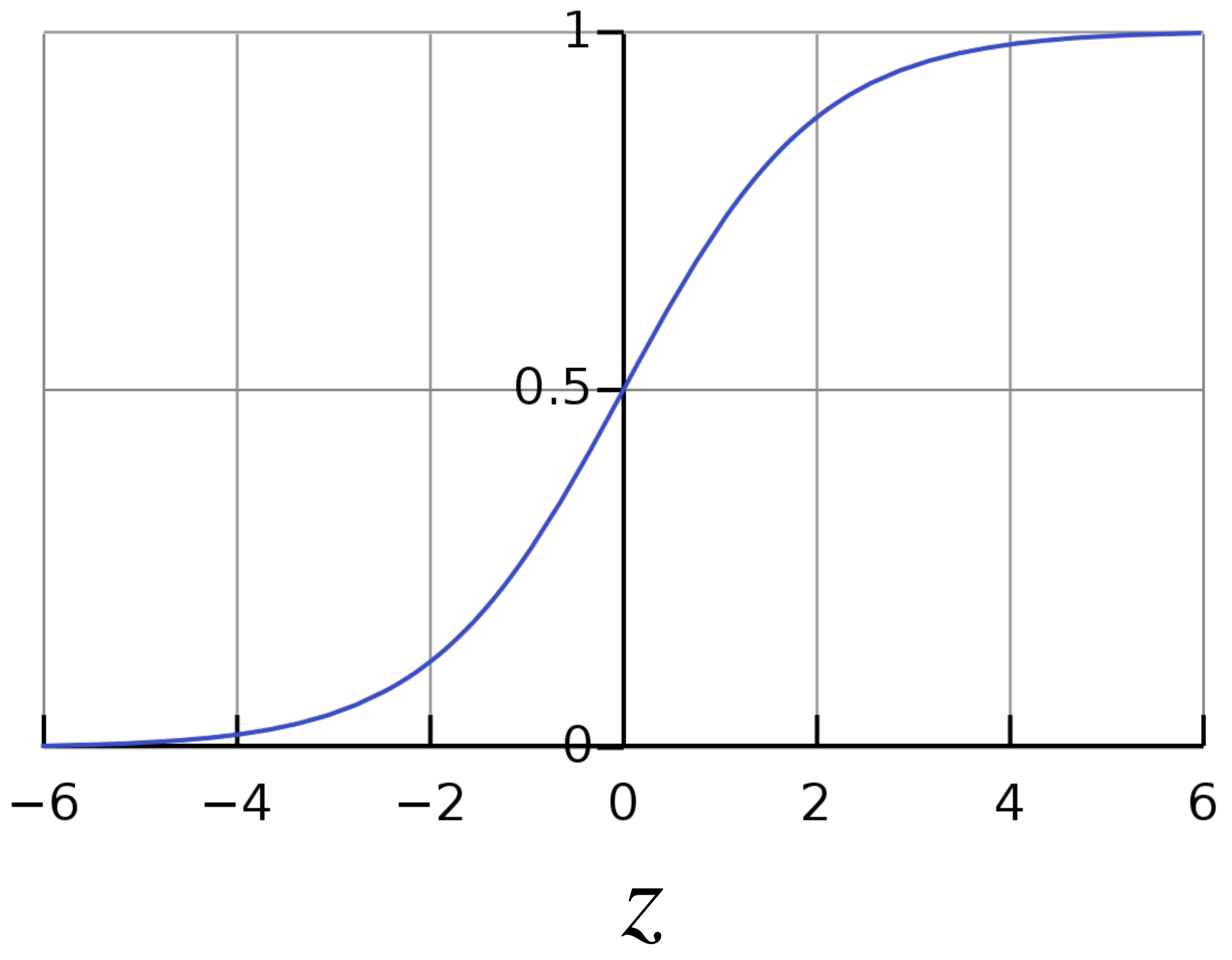
it	6
I	5
the	4
to	3
and	3
seen	2
yet	1
would	1
whimsical	1
times	1
sweet	1
satirical	1
adventure	1
genre	1
fairy	1
humor	1
have	1
great	1
…	…

Classification function

- *Given*: Input feature vector $[x_1, x_2, \ldots, x_d]$
- *Output*: $P(y = 1 \mid x)$ and $P(y = 0 \mid x)$ *(binary classification)*
- Use a *function,* $F : \mathbb{R}^{\mathrm{d}} \rightarrow [0,1]$ to model the probability $P(y \mid x)$
- Sigmoid:

$$P(y|x) = \frac{1}{1 + e^{-z}}$$

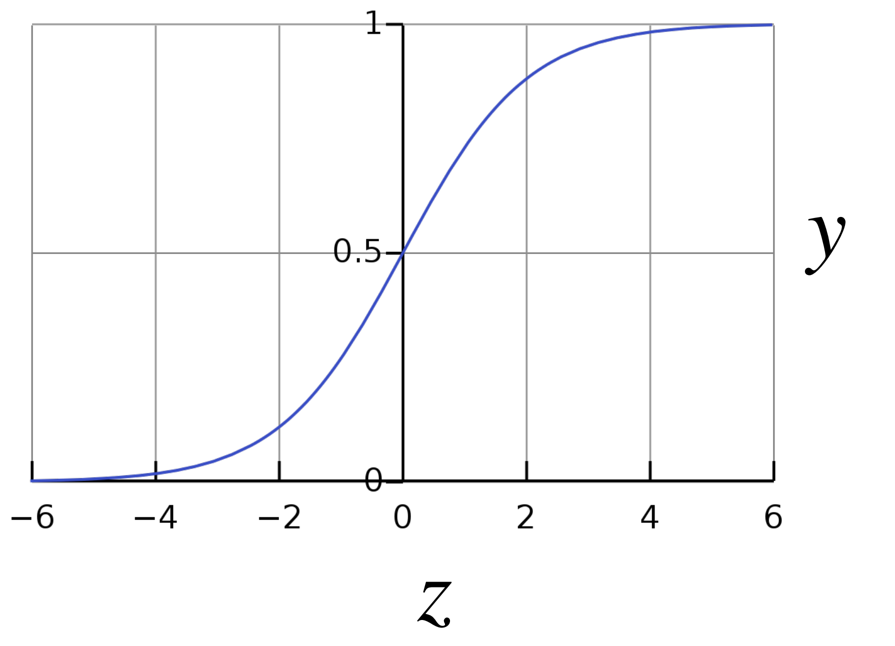
$$z = \sum_{i=1}^{d} w_i x_i + b$$



$$P(y|x)$$

Why the sigmoid?

$$y = \frac{1}{1+e^{-z}}$$



- Binary logistic regression uses the log-odds to determine the probability.

  - Log-odds:

$$\ell = \log \frac{p}{1-p}$$

Value is also known as logit (logistic unit)

$$\frac{p}{1-p} = \exp(\ell)$$

$$p = \exp(\ell)(1-p)$$

$$p(1+\exp(\ell)) = \exp(\ell)$$

$$p = \frac{\exp(\ell)}{(1+\exp(\ell))} = \frac{1}{(1+\exp(-\ell))} = \sigma(\ell)$$

- z is a linear estimate of the log-odds: $z = \sum_{i=1}^{d} w_i x_i + b$

Weights and bias

- *Which features are important* and *how much*?

- Learn a vector of **weights** and a **bias**

- Weights: Vector of real numbers, $\mathbf{w} = [w_1, w_2, \ldots, w_d]$

- Bias: Scalar intercept, b

$$z = \sum_{i=1}^{d} w_i x_i + b$$

Vector form

$$z = \mathbf{w} \cdot \mathbf{x} + b = \mathbf{w}^T \mathbf{x} + b$$

What is the bias?

- Let's say we have a feature that is always set to 1 regardless of what the input text is.
- This is clearly not an informative feature. However, let's say it was the only one I had…

first, how many weights do I need to learn for this feature?

(Credits: Richard Socher)

What is the bias?

- Let's say we have a feature that is always set to 1 regardless of what the input text is.
- This is clearly not an informative feature. However, let's say it was the only one I had…

$$\mathbf{w} \cdot \mathbf{x} + \boxed{b}$$

first, how many weights do I need to learn for this feature?

okay… what is the best set of weights for it?

(Credits: Richard Socher)

Putting it together

Given x, compute $z = \mathbf{w} \cdot \mathbf{x} + b$

Compute probabilities: $P(y = 1 \,|\, x) = \sigma(z) = \dfrac{1}{1 + e^{-z}}$

$$P(y = 1) = \sigma(\mathbf{w} \cdot \mathbf{x} + b)$$
$$\frac{1}{1 + e^{-(\mathbf{w} \cdot \mathbf{x} + b)}}$$

$$\begin{aligned} P(y = 0) &= 1 - \sigma(\mathbf{w} \cdot \mathbf{x} + b) \\ &= 1 - \frac{1}{1 + e^{-(\mathbf{w} \cdot \mathbf{x} + b)}} \\ &= \frac{e^{-(\mathbf{w} \cdot \mathbf{x} + b)}}{1 + e^{-(\mathbf{w} \cdot \mathbf{x} + b)}} \end{aligned}$$

Decision boundary:

$$\hat{y} = \begin{cases} 1 & \text{if } P(y = 1 \,|\, x) > 0.5 \\ 0 & \text{otherwise} \end{cases}$$

Decision boundary

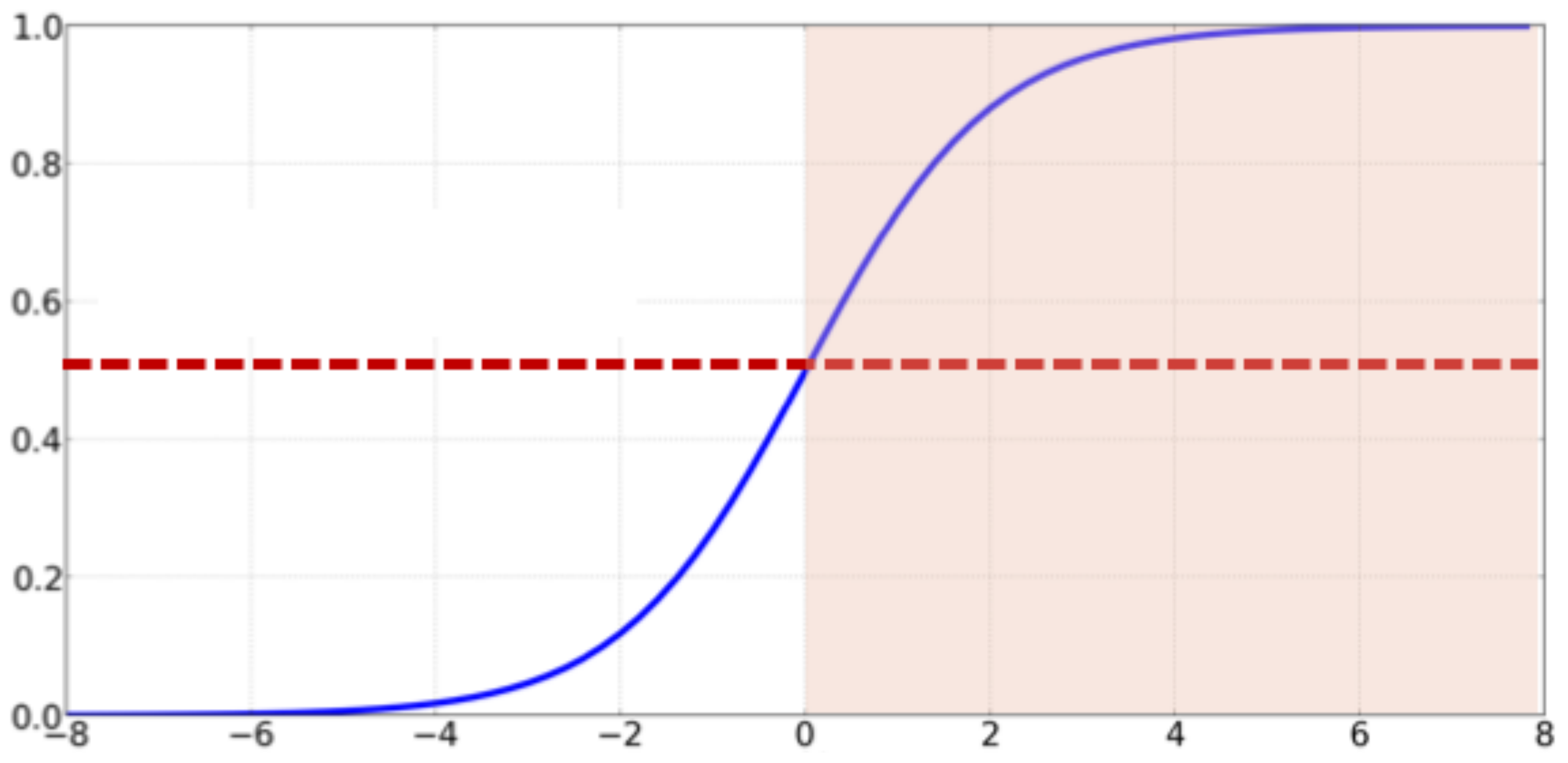
Threshold can be set as hyperparameter that is tuned

Decision boundary:

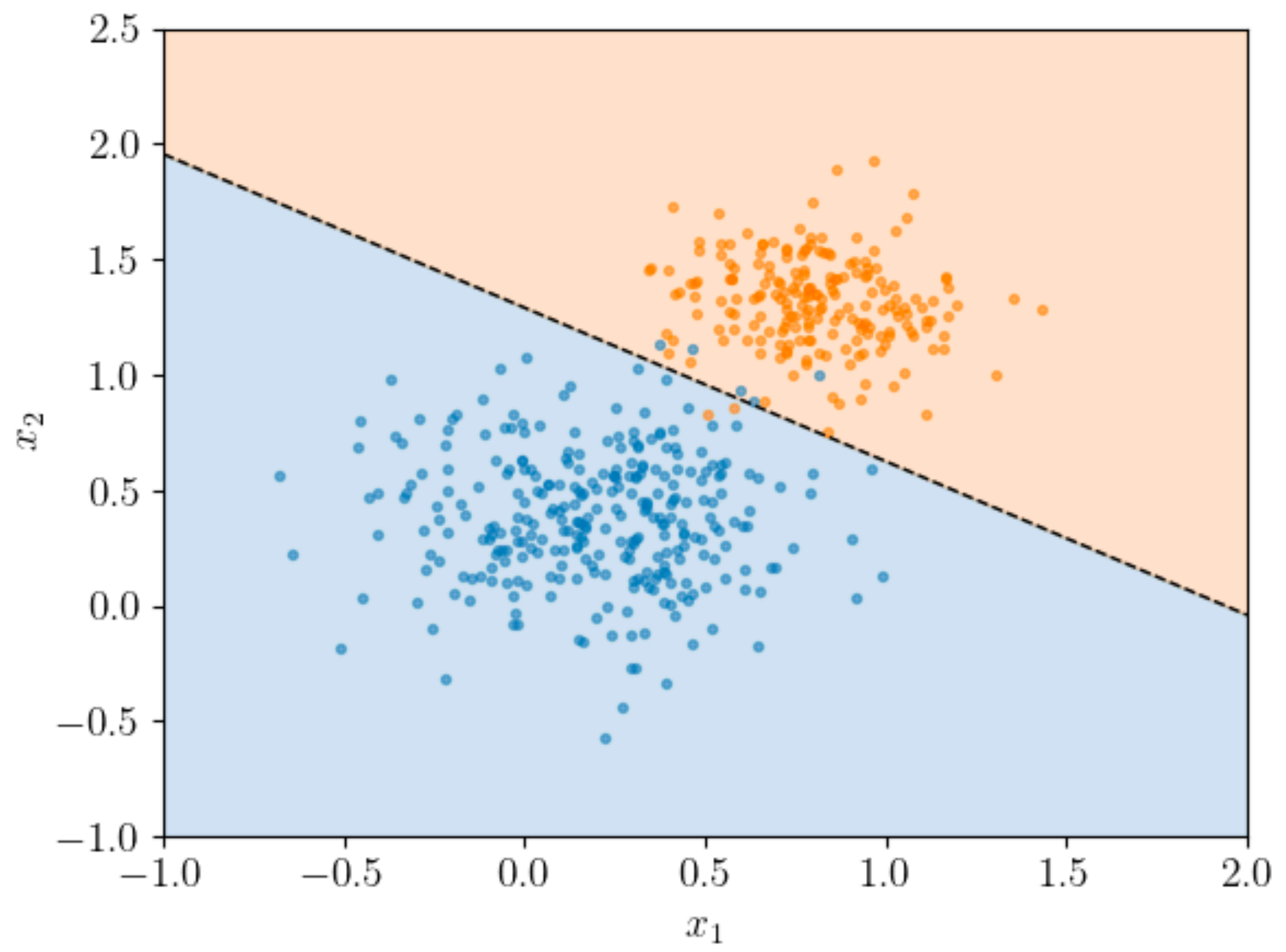
$$\hat{y} = \begin{cases} 1 & \text{if } P(y=1\,|\,x) > 0.5 \\ 0 & \text{otherwise} \end{cases}$$

$$\mathbf{w}\cdot\mathbf{x} + b > 0$$
$$\mathbf{w}\cdot\mathbf{x} + b \leq 0$$

$P(y\,|\,\mathbf{x})$



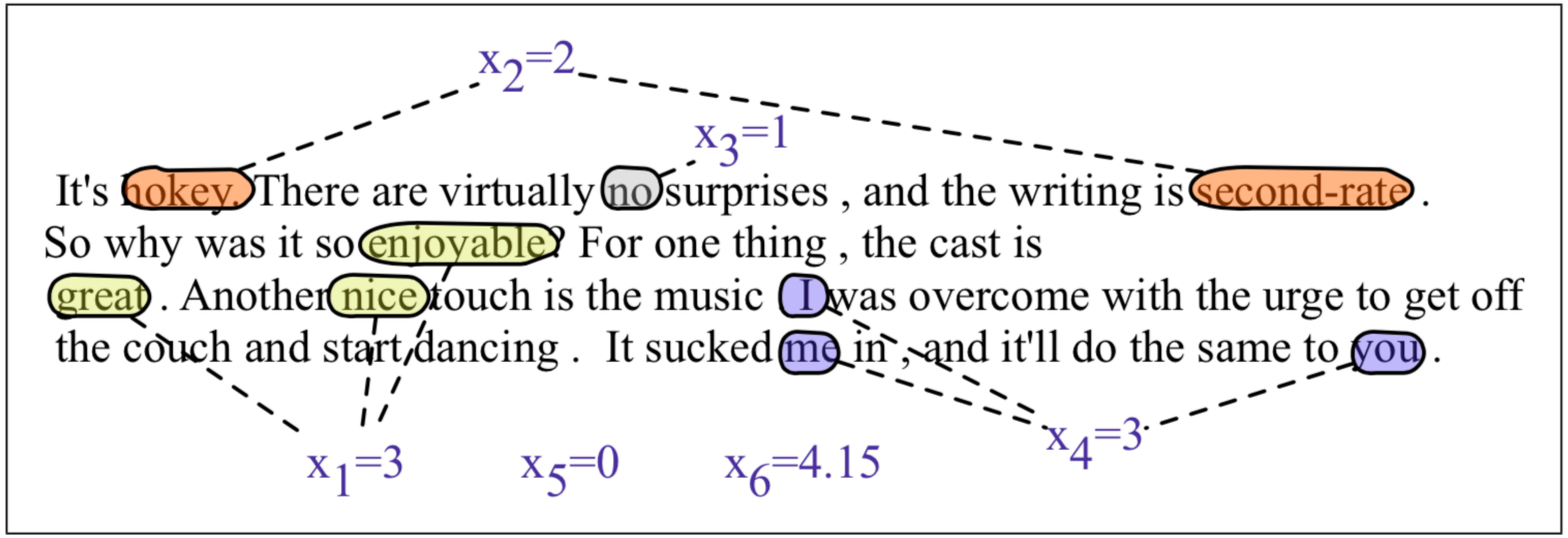
$\mathbf{w}\cdot\mathbf{x} + b$



Decision boundary is linear function of features

What do we use as features?

Example: Sentiment classification



Var	Definition	Value in Fig. 5.2
x_1	count(positive lexicon) $\in$ doc)	3
x_2	count(negative lexicon) $\in$ doc)	2
x_3	$\begin{cases} 1 & \text{if "no"} \in \text{doc} \\ 0 & \text{otherwise} \end{cases}$	1
x_4	count(1st and 2nd pronouns $\in$ doc)	3
x_5	$\begin{cases} 1 & \text{if "!"} \in \text{doc} \\ 0 & \text{otherwise} \end{cases}$	0
x_6	$\log(\text{word count of doc})$	$\ln(64) = 4.15$

(Credits: Dan Jurafsky)

Example: Sentiment classification

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x_6	log(word count of doc)	$\ln(64) = 4.15$

- Assume weights $w = [2.5, -5.0, -1.2, 0.5, 2.0, 0.7]$ and bias $b = 0.1$

$$\begin{aligned}
p(+|x) = P(Y=1|x) &= \sigma(w \cdot x + b) \\
&= \sigma([2.5, -5.0, -1.2, 0.5, 2.0, 0.7] \cdot [3, 2, 1, 3, 0, 4.15] + 0.1) \\
&= \sigma(.805) \\
&= 0.69 \\
p(-|x) = P(Y=0|x) &= 1 - \sigma(w \cdot x + b) \\
&= 0.31
\end{aligned}$$

Feature design

- Most important rule: Data is *key*!

- Linguistic intuition (e.g. part of speech tags, parse trees)

- Complex combinations

$$x_1 = \begin{cases} 1 & \text{if "}\mathit{Case}(w_i) = \text{Lower"} \\ 0 & \text{otherwise} \end{cases}$$

$$x_2 = \begin{cases} 1 & \text{if "}w_i \in \text{AcronymDict"} \\ 0 & \text{otherwise} \end{cases}$$

$$x_3 = \begin{cases} 1 & \text{if "}w_i = \text{St. \& }\mathit{Case}(w_{i-1}) = \text{Cap"} \\ 0 & \text{otherwise} \end{cases}$$

- Feature templates

  - Sparse representations, hash only seen features into index

  - Ex. Trigram(*logistic regression classifier*) = Feature #78

- Advanced: Representation learning (we will see this with neural networks!)

Logistic Regression: what’s good and what’s not

- More freedom in designing features
  - No strong independence assumptions like Naive Bayes
  - More robust to correlated features (“San Francisco” vs “Boston”) —LR is likely to work better than NB
  - Can even have the same feature twice! (*why?*)
- **However**: not as good on small datasets (compared to Naive Bayes)
- Interpreting learned weights can be challenging

How to learn the parameters?

Learning

- We have our **classification function -** how to assign weights and bias?

- Goal: predicted label $\hat{y}$ as close as possible to actual label y

  - **Distance metric/Loss function** between $\hat{y}$ and y :
    $L(\hat{y}, y)$

  - **Optimization algorithm** for updating weights

Loss function

- Assume $\hat{y} = \sigma(\mathbf{w} \cdot \mathbf{x} + b)$
- $L(\hat{y}, y) =$ Measure of difference between $\hat{y}$ and y. But what form?
- Maximum likelihood estimation (conditional):
    - Choose $\mathbf{w}$ and b such that $\log P(y \mid x)$ is maximized for true labels y paired with input x
    - Similar to language models!
        - $\max \log P(w_t \mid w_{t-n}, \ldots, w_{t-1})$ given a corpus

Cross Entropy loss

- Assume a single data point (x, y) and two classes

- Classifier probability: $P(y\,|\,x) = \hat{y}^{\,y}(1 - \hat{y})^{1-y}$

- Log probability: $\log P(y\,|\,x) = \log[\hat{y}^{\,y}(1 - \hat{y})^{1-y}]$

$$= y\log\hat{y} + (1 - y)\log(1 - \hat{y})$$

- Loss: $-\log P(y\,|\,x) = -\,[y\log\hat{y} + (1 - y)\log(1 - \hat{y})]$

$y = 1 \Rightarrow -\log\hat{y}$ $\qquad$ $y = 0 \Rightarrow -\log(1 - \hat{y})$

Cross Entropy loss

- Assume n data points $(x^{(i)}, y^{(i)})$
- Classifier probability: $\prod_{i=1}^{n} P(y^{(i)} \mid x^{(i)}) = \prod_{i=1}^{n} \hat{y}^{y}(1-\hat{y})^{1-y}$
- Loss: $-\log \prod_{i=1}^{n} P(y^{(i)} \mid x^{(i)}) = -\sum_{i=1}^{n} \log P(y^{(i)} \mid x^{(i)})$

$$L_{CE} = -\sum_{i=1}^{n} [y \log \hat{y} + (1-y)\log(1-\hat{y})]$$

Maximizing likelihood = Minimizing cross entropy

Example: Computing CE Loss

Var	Definition	Va
x_1	count(positive lexicon) $\in$ doc)	3
x_2	count(negative lexicon) $\in$ doc)	2
x_3	$\begin{cases} 1 & \text{if “no”} \in \text{doc} \\ 0 & \text{otherwise} \end{cases}$	1
x_4	count(1st and 2nd pronouns $\in$ doc)	3
x_5	$\begin{cases} 1 & \text{if “!”} \in \text{doc} \\ 0 & \text{otherwise} \end{cases}$	0
x_6	log(word count of doc)	$\ln(64) = 4.15$

- Assume weights $w = [2.5, -5.0, -1.2, 0.5, 2.0, 0.7]$ and bias $b = 0.1$
- If y = 1 (positive sentiment), $L_{CE} = -\log(0.69) = 0.37$
- If y = 0 (negative sentiment), $L_{CE} = -\log(0.31) = 1.17$

Properties of CE Loss

- $L_{CE} = -\sum_{i=1}^{n} [y^{(i)} \log \hat{y}^{(i)} + (1 - y^{(i)}) \log(1 - \hat{y}^{(i)})]$

- Ranges from 0 (perfect predictions) to ∞

- Lower the value, better the classifier

- *Cross-entropy* between the true distribution $P(y \mid x)$ and predicted distribution $P(\hat{y} \mid x)$

$$L_{CE} = -\frac{n_{y=1}}{n} \log \hat{y} - \frac{n_{y=0}}{n} \log(1 - \hat{y}) = \sum_{y=0}^{1} p(y \mid x) \log p(\hat{y} \mid x)$$

Optimization

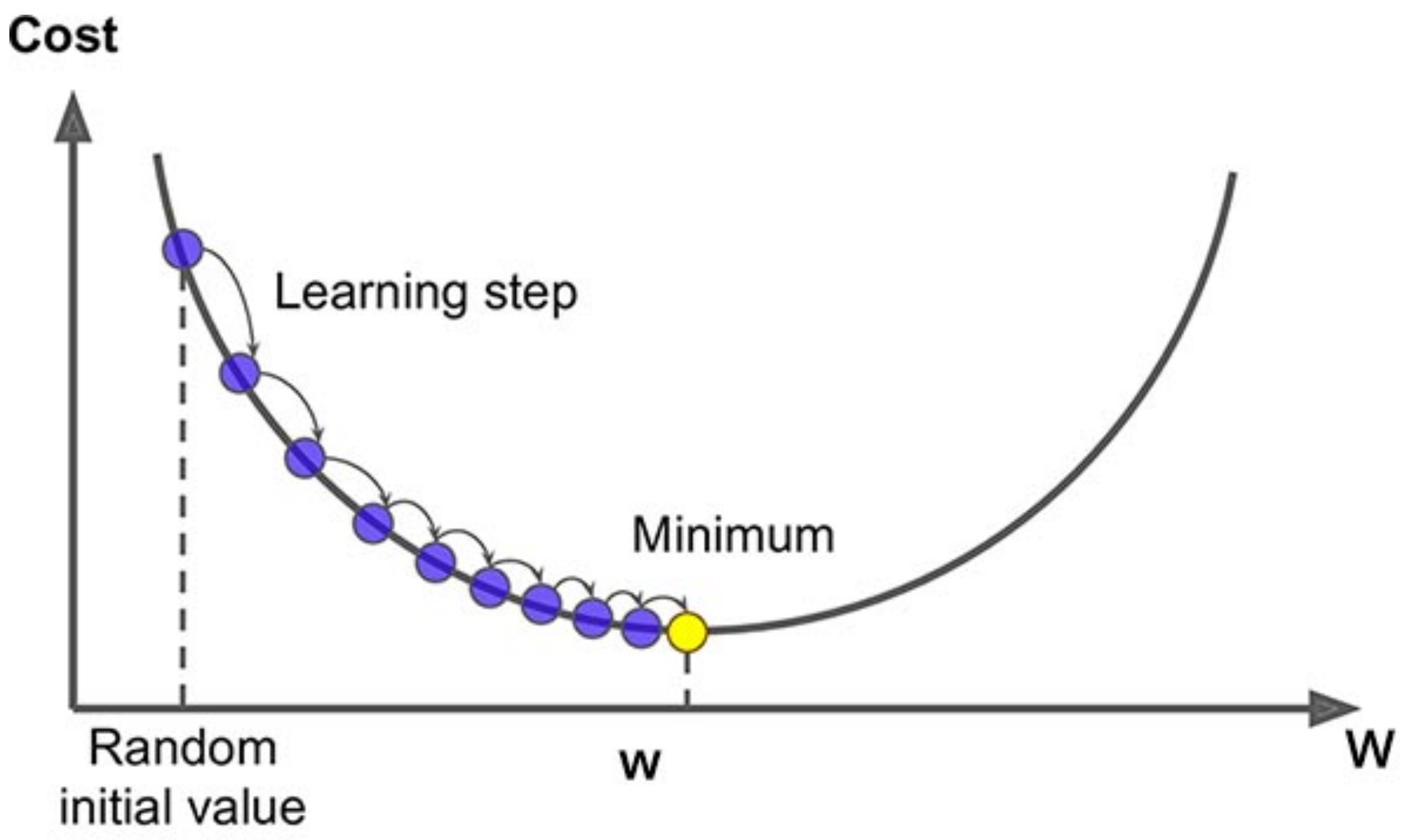
- We have our **classification function** and **loss function** - how do we find the best w and b?

$$\theta = [w; b]$$

$$\hat{\theta} = \arg\min_{\theta} \frac{1}{n} \sum_{i=1}^{n} L_{CE}(y^{(i)}, x^{(i)}; \theta)$$

- No close form solution: need to numerically determine optimal solution
- Gradient descent:
  - Find direction of steepest slope
  - Move in the opposite direction

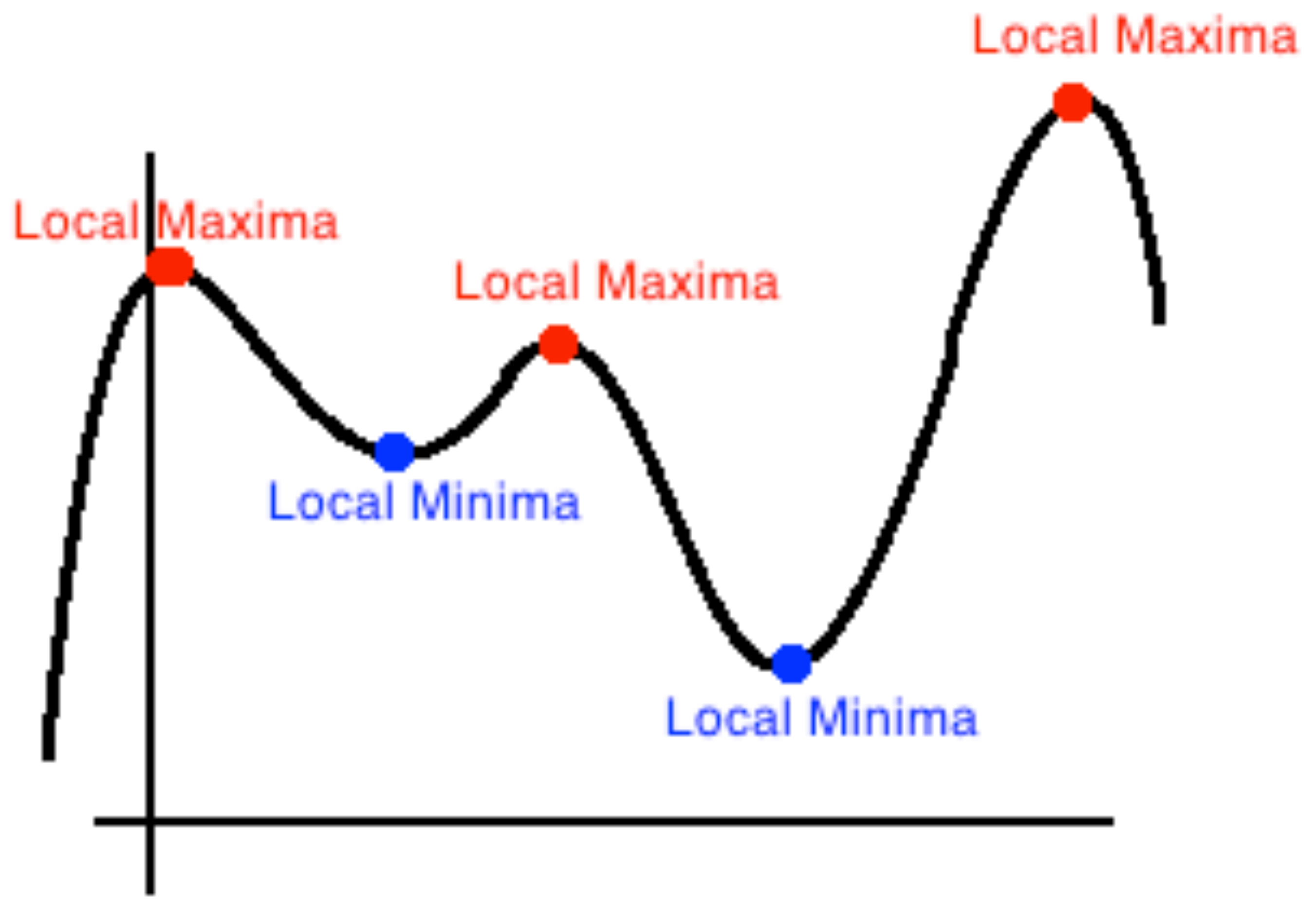
Gradient descent (1-D)



$$\theta^{t+1} = \theta^{t} - \eta \frac{d}{d\theta} f(x;\theta)$$

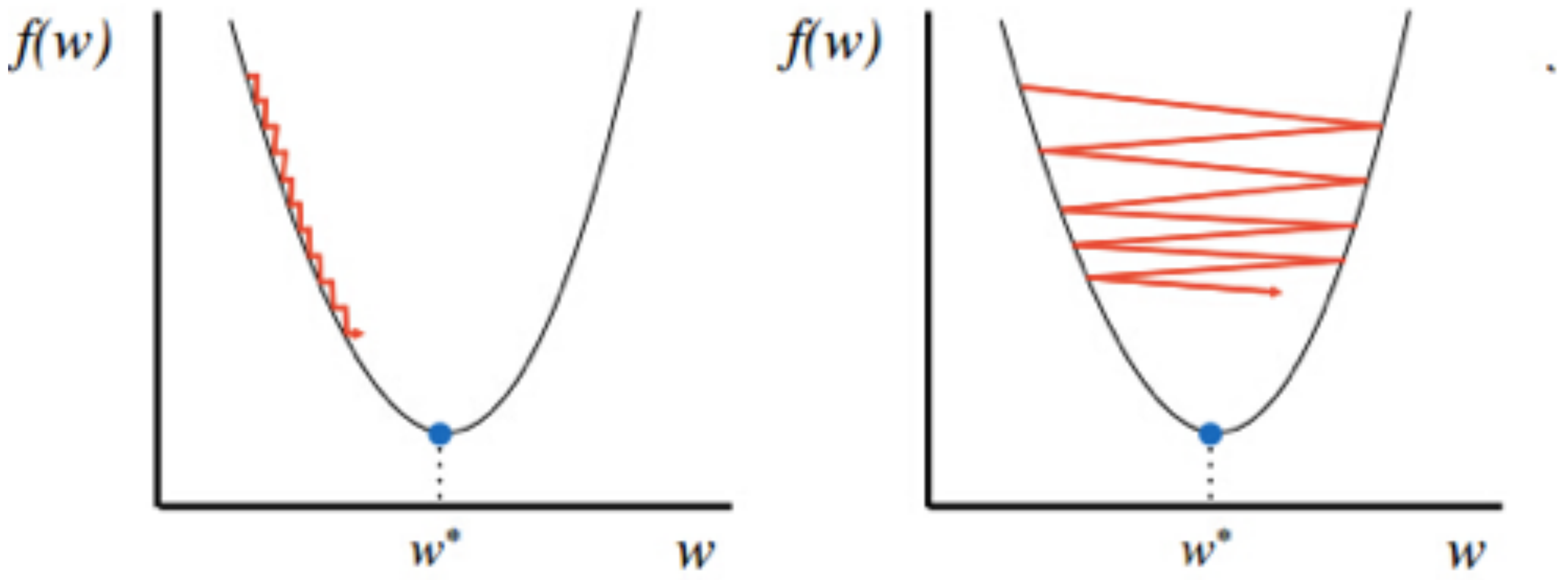
Gradient descent for LR

- Cross entropy loss for logistic regression is **convex** (i.e. has only one global minimum)
  - No local minima to get stuck in
- Deep neural networks are not so easy
  - Non-convex



Learning Rate

- Updates: $\theta^{t+1} = \theta^t - \eta \frac{d}{d\theta} f(x;\theta)$
- Magnitude of movement along gradient
- Higher/faster learning rate = larger updates to parameters



Too small: converge very slowly

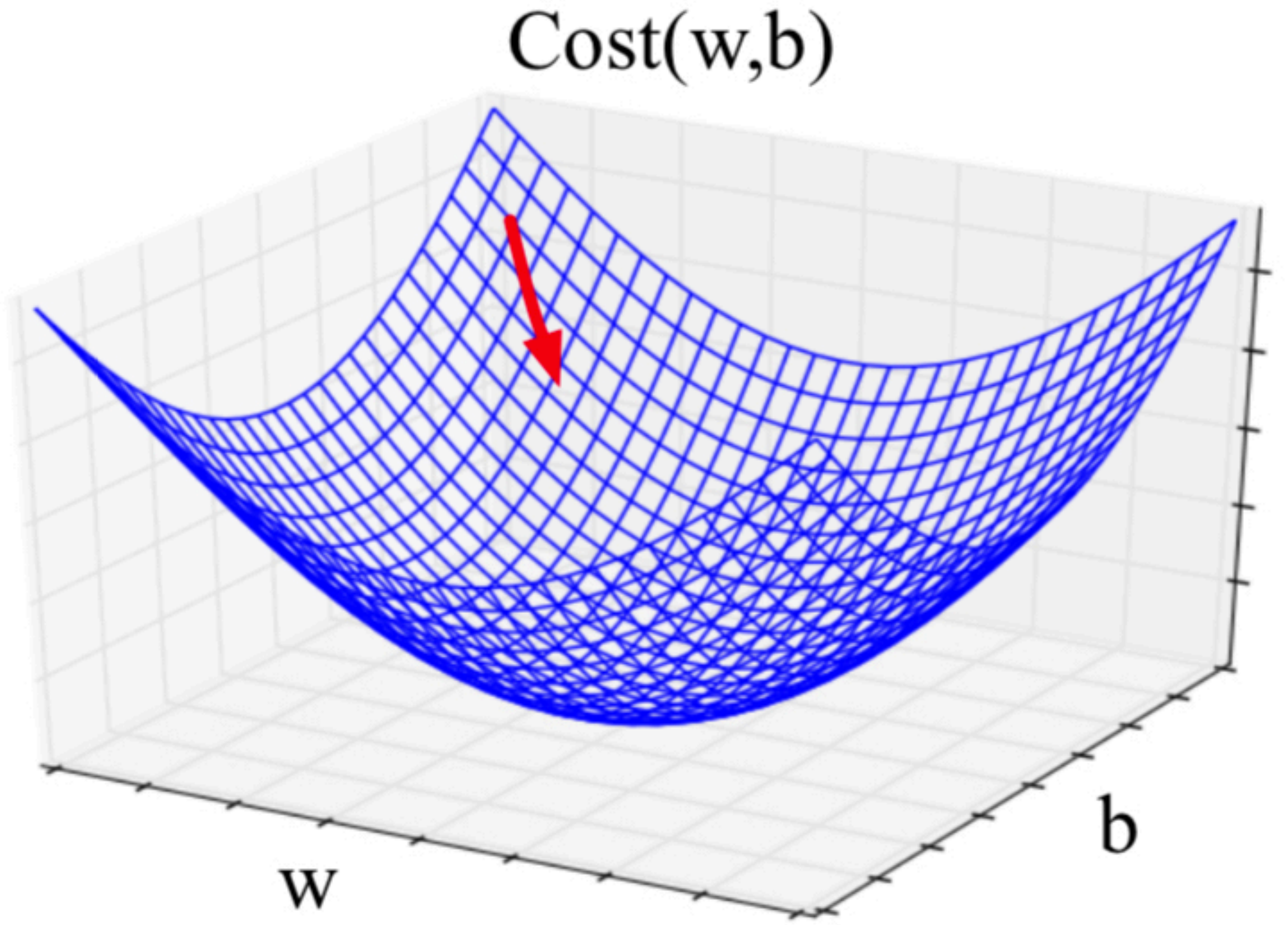
Too big: overshoot and even diverge

Gradient descent with vector weights

- In LR: weight w is a vector

- Express slope as a partial derivative of loss w.r.t each weight:

$$\nabla_{\theta} L(f(x;\theta),y)) = \begin{bmatrix} \frac{\partial}{\partial w_1} L(f(x;\theta),y) \\ \frac{\partial}{\partial w_2} L(f(x;\theta),y) \\ \vdots \\ \frac{\partial}{\partial w_n} L(f(x;\theta),y) \end{bmatrix}$$



- Updates: $\theta^{(t+1)} = \theta^{t} - \eta \nabla L(f(x;\theta), y)$

Gradient for logistic regression

$$L_{CE} = -\sum_{i=1}^{n} [y^{(i)} \log \sigma(w \cdot x^{(i)} + b) + (1 - y^{(i)}) \log(1 - \sigma(w \cdot x^{(i)} + b))]$$

Gradient, $$\frac{dL_{CE}(w,b)}{dw_j} = \sum_{i=1}^{n} [\underbrace{\sigma(w \cdot x^{(i)} + b) - y^{(i)}}_{\text{Diff between true y and prediction}}] x_j^{(i)}$$

input feature value

$$\frac{dL_{CE}(w,b)}{db} = \sum_{i=1}^{n} [\sigma(w \cdot x^{(i)} + b) - y^{(i)}]$$

Stochastic Gradient Descent

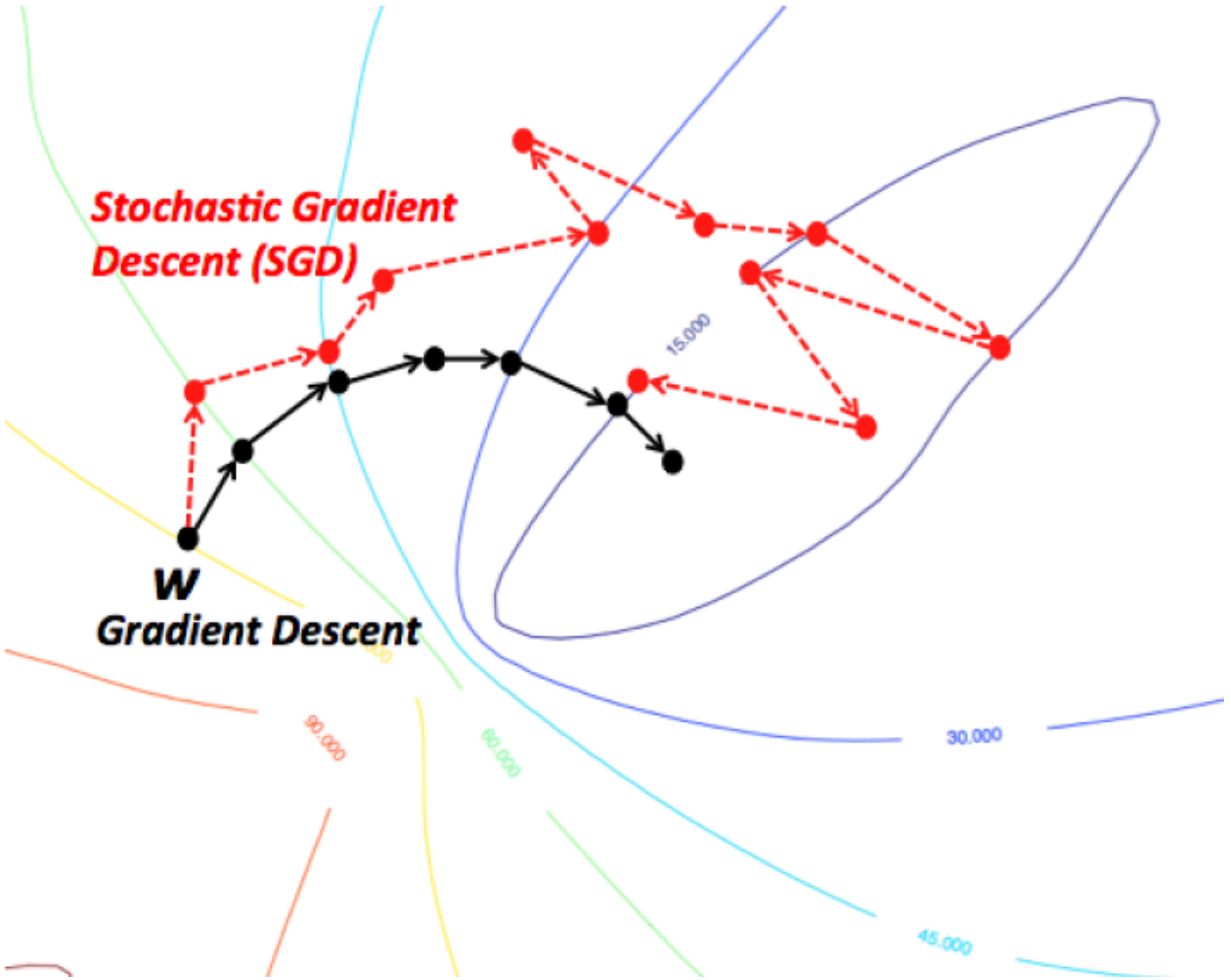
- Online optimization
- Compute loss and minimize after each training example

function STOCHASTIC GRADIENT DESCENT($L()$, $f()$, x, y) **returns** θ
    # where: L is the loss function
    #   f is a function parameterized by θ
    #   x is the set of training inputs $x^{(1)}, x^{(2)}, \ldots, x^{(n)}$
    #   y is the set of training outputs (labels) $y^{(1)}, y^{(2)}, \ldots, y^{(n)}$

$\theta \leftarrow 0$
repeat til done   # see caption
  For each training tuple $(x^{(i)}, y^{(i)})$ (in random order)
    1. Optional (for reporting):   # How are we doing on this tuple?
Per Instance Loss
      Compute $\hat{y}^{(i)} = f(x^{(i)};\theta)$ # What is our estimated output $\hat{y}$?
      Compute the loss $L(\hat{y}^{(i)}, y^{(i)})$ # How far off is $\hat{y}^{(i)}$) from the true output $y^{(i)}$?
    2. $g \leftarrow \nabla_\theta L(f(x^{(i)};\theta), y^{(i)})$ # How should we move θ to maximize loss?
    3. $\theta \leftarrow \theta - \eta\, g$ # Go the other way instead
return θ

Stochastic Gradient Descent

- Online optimization
- Compute loss and minimize after each training example



Regularization

- Training objective: $\hat{\theta} = \arg\max_{\theta} \sum_{i=1}^{n} \log P(y^{(i)} \mid x^{(i)})$

- This might fit the training set too well! (including noisy features)

- Poor generalization to the unseen test set — **Overfitting**

- **Regularization** helps prevent overfitting

$$\hat{\theta} = \arg\max_{\theta} \sum_{i=1}^{n} \log P(y^{(i)} \mid x^{(i)}) \overbrace{- \alpha R(\theta)}^{\text{Penalize large weights}}$$

L2 regularization

- $R(\theta) = ||\theta||^2 = \sum_{j=1}^{d} \theta_j^2$

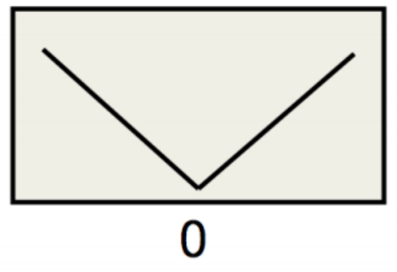
- Euclidean distance of weight vector θ from origin

- L2 regularized objective:

$$\hat{\theta} = \arg\max_{\theta} \sum_{i=1}^{n} \log P(y^{(i)} | x^{(i)}) - \alpha \sum_{j=1}^{d} \theta_j^2$$

L1 Regularization

- $R(\theta) = ||\theta||_1 = \sum_{j=1}^{d} |\theta_j|$



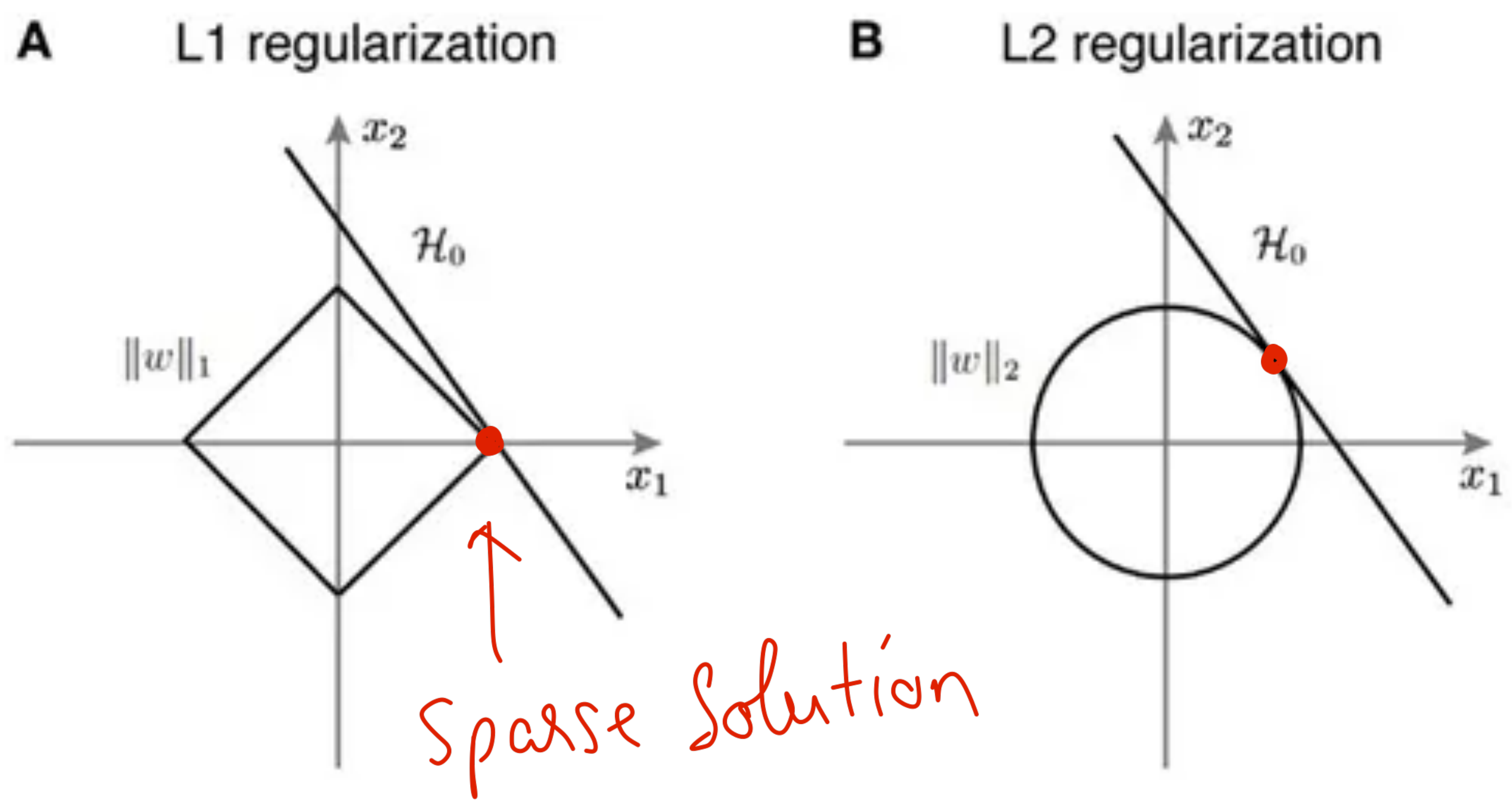
- Manhattan distance of weight vector θ from origin

- L1 regularized objective:

$$\hat{\theta} = \arg\max_{\theta} \sum_{i=1}^{n} \log P(y^{(i)} | x^{(i)}) - \alpha \sum_{j=1}^{d} |\theta_j|$$

L2 vs L1 regularization

- L2 is easier to optimize - simpler derivation
  - L1 is complex since the derivative of $|\theta|$ is not continuous at 0



- L2 leads to many small weights (due to θ^2 term)
  - L1 prefers *sparse* weight vectors with many weights set to 0 (i.e. far fewer features used)

Multinomial Logistic Regression

Multinomial Logistic Regression

- What if we have more than 2 classes? (e.g. Part of speech tagging, named entity recognition)
- Need to model $P(y = c \mid x) \forall c \in C$
- Generalize **sigmoid** function to **softmax**

$$\text{softmax}(z_i) = \frac{e^{z_i}}{\sum_{j=1}^{k} e^{z_j}} \quad 1 \le i \le k$$

Normalization

Softmax

- Similar to sigmoid, softmax squashes values towards 0 or 1
- If $z = [0,1,2,3,4]$, then
  - softmax$(z) = ([0.0117,0.0317,0.0861,0.2341,0.6364])$
- For multinomial LR,

$$P(y = c \,|\, x) = \frac{e^{w_c \cdot x + b_c}}{\sum_{j=1}^{k} e^{w_j \cdot x + b_j}}$$

Features in multinomial LR

- Features need to include both input (x) and class (c)

- Implicit in binary case

Var	Definition	Wt
$f_1(0,x)$	$\begin{cases} 1 & \text{if "!"} \in \text{doc} \\ 0 & \text{otherwise} \end{cases}$	-4.5
$f_1(+,x)$	$\begin{cases} 1 & \text{if "!"} \in \text{doc} \\ 0 & \text{otherwise} \end{cases}$	2.6
$f_1(-,x)$	$\begin{cases} 1 & \text{if "!"} \in \text{doc} \\ 0 & \text{otherwise} \end{cases}$	1.3

Learning

- Generalize binary loss to multinomial CE loss:

$$L_{CE}(\hat{y}, y) = -\sum_{c=1}^{k} 1\{y = k\} \log P(y = k \,|\, x)$$

$$= -\sum_{c=1}^{k} 1\{y = k\} \log \frac{e^{w_c \cdot x + b_c}}{\sum_{j=1}^{k} e^{w_j \cdot x + b_c}}$$

Binary CE loss

$$-\sum_{i=1}^{n} [y \log \hat{y} + (1 - y) \log(1 - \hat{y})]$$

- Gradient:

$$\frac{dL_{CE}}{dw_c} = -(1\{y = c\} - P(y = c \,|\, x)) x_c$$

$$= -\left(1\{y = c\} - \frac{e^{w_c \cdot x + b_c}}{\sum_{j=1}^{k} e^{w_j \cdot x + b_c}}\right) x_c$$